ROUND 1 REVIEW

AI Frontier Challenge 2026 Round 1 Submission

Team Name

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Track

H5. Frontline Health-Worker Companion for Low-Connectivity Areas

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Use Case

Problem Statement

- **Frontline Overload & Zero Mobile Reception:**
  India's 10.29 lakh ASHAs and 2.5 lakh ANMs manage up to 28 paper registers [5]; in Andhra Pradesh alone, 1,700+ remote villages (1,000+ in ASR district) have complete zero mobile reception [4].
- **Preventable Maternal & Neonatal Mortality:**
  Obstetric haemorrhage causes 47% of maternal deaths [3]; unrecognised newborn danger signs (hypothermia, convulsions, chest indrawing) lead to fatal referral delays [1, 2].
- **AI Safety & Frontline Malpractice Risks:**
  General-purpose AI can produce inaccurate or unsafe medical guidance without appropriate safeguards, risking non-prescribing ASHA boundary violations [1], failing offline, and lacking WHO clinical guardrails [6].

Market Opportunity

- **Frontline Scale & National Need:**
  10.29 lakh ASHAs and ~2.5 lakh Sub-Centre ANMs across India serving 1,000 population norms, requiring an intuitive, reliable paramedical companion [1, 2].
- **Critical Gap in Existing Systems:**
  Existing state tools (ANMOL, Kilkari) focus on administrative reporting or outbound audio broadcasts; they rely on mobile networks [4] and lack on-demand offline clinical danger triage or strict scope gating [5].
- **Sahaaya High-Impact Value:**
  100% offline, protocol-grounded companion: deterministic danger-sign escalation [1], physician redirects [6], prioritized visits, emergency referral slips, and 2G SMS field sync [4].

Official Protocol Sources & Clinical Evidence Citations:

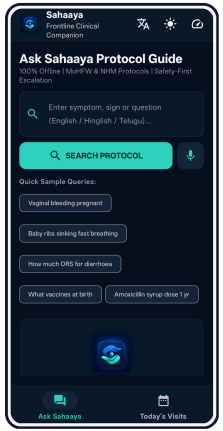
[1] MoHFW National Guidelines: Revised India MCP Card (2018) & ASHA Modules 6-7
[2] NHM Clinical Guidance: ASHA Induction Module (pp. 51-97) & STOP Diarrhoea
[3] PMC9292773 Study: Obstetric haemorrhage causes 47% maternal deaths in India
[4] TelecomTalk AP Survey: 1,700 remote hamlets (ASR dist) zero cellular signal
[5] New Lines / IJCMPH 2024: Frontline overload (28 paper registers, 23 apps)
[6] WHO AI Safety Standards: Multi-modal AI guidance for clinical decision support

Proposed Solution

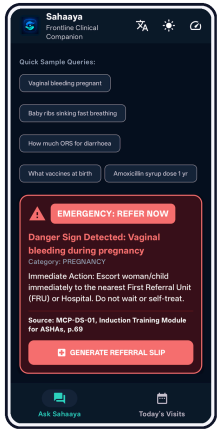
Solution Overview

- **Deterministic Danger Escalation + Referral Slip:**
  Scans queries across English, Hinglish & Telugu-Latin; triggers red screen (100% recall on 367 phrasings [1, 2]) + 1-tap 108 Emergency Referral Slip.
- **Strict Non-Prescribing Scope Gate:**
  Hard-blocks antibiotic & dosage questions with physician redirects, ensuring paramedical workers operate strictly within ASHA guidelines [1, 6].
- **Extractive BM25 Protocol Retrieval:**
  Pure Kotlin BM25 matches questions to official MoHFW/NHM manuals [1, 2] with exact document & page citations (0.53 ms latency, 0% hallucination).
- **Prioritised Daily Household Action Plan & 2G Sync:**
  Triage ranking: Danger Flags > Overdue Follow-ups > Routine Due Dates + 1-Tap End-of-Day PHC Summary Report (transmissible via 2G SMS [4, 5]).

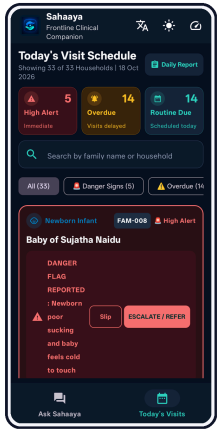
SCREEN 1: SEARCH & VOICE
Offline Trilingual Voice & Text



SCREEN 2: SAFETY ESCALATION
100% Danger Recall + 108 Slip



SCREEN 3: TODAY'S VISITS
Urgency Triage & Action Plan

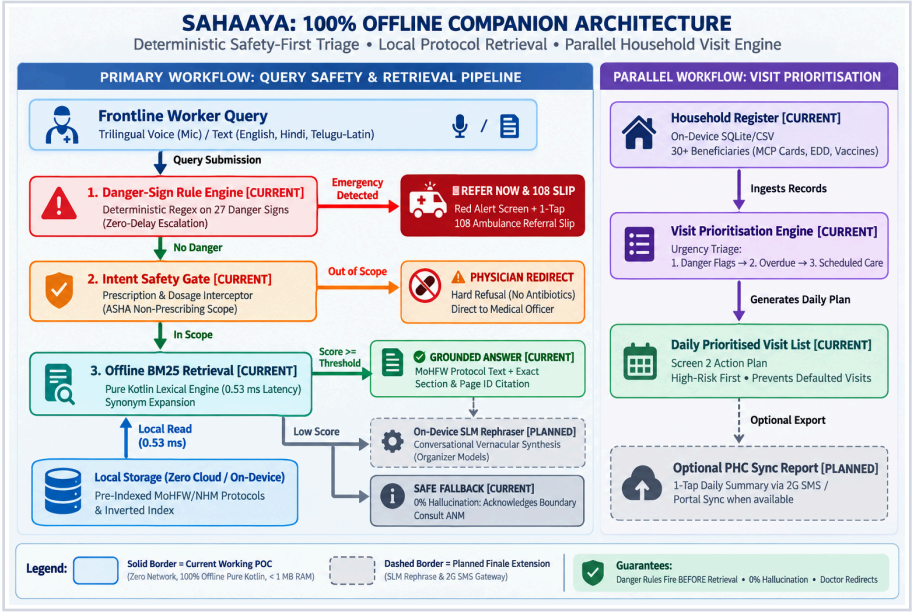


Design Steps & Architecture

Technical Implementation

1. **Input & Speech Recognition:**
   Trilingual voice mic and text (EN, Hindi, Telugu-Latin) for instant frontline accessibility.
2. **Danger-Sign Rule Engine [Current]:**
   Deterministic regex on 27 danger signs (100% recall [1, 2]); immediately triggers Red Alert + 108 Referral Slip.
3. **Intent Safety Gate [Current]:**
   Intercepts dosage & prescription queries; enforces non-prescribing boundaries [1, 6] with physician redirects.
4. **Offline BM25 Retrieval [Current]:**
   Pure Kotlin search over local on-device MoHFW/NHM protocol chunks; exact page citations (0% hallucination).
5. **Parallel Household Engine [Current]:**
   Ingests offline household register -> Urgency triage (Danger > Overdue > Scheduled) -> Daily Prioritised Visit List.
6. **Planned Extensions [Roadmap]:**
   On-device SLM rephraser for natural conversational synthesis; automated 2G SMS field summary sync.

Architecture Diagram (Safety-First & Parallel Workflows)



Challenges & Mitigation

Challenge 1

Problem: Zero mobile reception in rural hamlets and severe RAM/battery constraints on frontline smartphones.

Mitigation: 100% on-device architecture with zero network permissions. Measured across 30 diagnostic queries on physical hardware (OnePlus CPH2613, Android 15 in Airplane Mode) and standalone JVM: APK size is 16.9 MB, median query latency is 0.53 ms (P95: 5.38 ms), and peak heap RAM footprint is < 1 MB.

Challenge 2

Problem: Medical hallucination risks and catastrophic delays from missed pregnancy and newborn danger signs.

Mitigation: Deterministic danger rules execute BEFORE any retrieval logic. Tested across a 367-phrase benchmark derived from 27 MoHFW/NHM emergency signs across English, Hinglish & Telugu-Latin colloquial variants: achieved 100% recall on the test set, triggering instant Red Alerts + 108 referral slips without retrieval delay.

Challenge 3

Problem: Code-mixed vernacular queries (English, Hinglish, Telugu-Latin) with dialectal variations, typos, and paper register friction.

Mitigation: [Implemented]: Robust phonetic tokenisation, code-mixed synonym mapping, and clinical urgency triage (Danger > Overdue > Scheduled) with 1-tap referral slips. [Planned]: On-device SLM rephraser using pre-cached organizer models and background 2G SMS gateway sync for daily PHC reporting.

Unique Selling Proposition (USP)

Key Differentiator 1

Deterministic Safety Pipeline:
Emergency rules execute BEFORE any AI logic; 100% danger recall on 367 benchmark phrasings [1, 2]. Hard non-prescribing gate [6] with extractive, source-cited responses & 108 slips.

Key Differentiator 2

100% Offline & Measured Speed:
Zero network permissions (built for [4]).
OnePlus CPH2613 (Airplane Mode):
· Median Query Latency: 0.53 ms
· P95 Benchmark Latency: 5.38 ms
· APK: 16.9 MB | Peak RAM: < 1 MB
Runs natively without SIM or signal.

Key Differentiator 3

Prioritised Household Action Plan:
Transforms 30+ static paper records into an actionable daily visit list.
Ensures high-risk pregnant women & unimmunized infants are visited first (Danger > Overdue > Scheduled).
[Planned Extension: 2G SMS sync].

Competitive Advantage Summary

· **Integrated Frontline Triad:**
Sahaaya uniquely unifies three essential frontline capabilities in one zero-network app: deterministic emergency danger escalation [1], offline MoHFW/NHM protocol retrieval with page citations [2], and clinical urgency-ranked household visit planning.

· **Clinically Safe & Field-Hardened:**
Guarantees paramedical boundary safety via hard physician redirects on prescriptions [1, 6], operates 100% offline in zero-connectivity hamlets [4], and delivers instant 1-tap referral slips to eliminate fatal transport delays [3].

Demo & POC Video

Live Demonstration

YouTube Demo: https://youtu.be/_Yz6aoMcbP4 **GitHub:** github.com/yogeswar142/Sahaaya **APK:** v1.0.0 (16.9 MB)

Working Offline Android POC | Grounded on MoHFW/NHM Protocols [1-3] | 100% Danger Recall | Zero Cloud Permissions